# Math is Playful — Homework

*Lesson 10-3 • Standard MPP.7*

**Name:** _____ **Period:** _____ **Date:** _____

## Today's Goals

Content: I can use patterns and relationships to make sense of the Tower of Hanoi puzzle and think logically about its solutions.

Language: I can describe a puzzle strategy using the words pattern, rule, and predict.

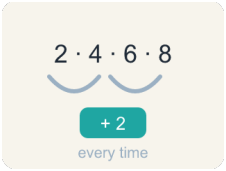
## Key Words



**puzzle (rompecabezas)** — A problem designed to be played with — it has rules, a goal, and room to imagine solutions.



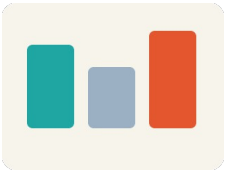

**pattern (patrón)** — Something that repeats or changes in a predictable way.




**pattern rule (regla del patrón)** — A rule that tells you how to get the next value in a pattern from the one before it.




**predict (predecir)** — To say what will happen before it happens, using a pattern or reasoning.



**organize (organizar)** — To arrange information — often in a table — so patterns become easier to see.

## Practice

*Show your work. Use the blank space under each problem.*

**1. What is the fewest number of moves needed to solve a three-disc Tower of Hanoi puzzle?**

A. 3 moves
B. 6 moves
C. 7 moves
D. 9 moves

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**2. Which move is against the rules of the Tower of Hanoi?**

A. Moving one disc at a time
B. Moving the top disc from a stack
C. Placing a larger disc on top of a smaller disc
D. Placing a smaller disc on top of a larger disc

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**3. The table shows 1 disc → 1 step, 2 discs → 3 steps, 3 discs → 7 steps. What is the pattern rule for the number of steps when you add one more disc?**

A. Add 2 to the number of steps
B. Add the number of discs to the steps
C. Multiply the number of steps by 3
D. Double the number of steps and add 1

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**4. Use the pattern rule to complete the table: how many steps for a four-disc tower?**

A. 9 steps
B. 14 steps
C. 15 steps
D. 28 steps

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**5. What is the minimum number of steps to solve a FIVE-disc Tower of Hanoi puzzle?**

A. 15 steps
B. 30 steps
C. 31 steps
D. 35 steps

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**6. How can you ORGANIZE the information about discs and moves to help you see patterns?**

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**7. Why is finding a pattern MORE powerful than just solving the three-disc puzzle by hand?**

A. The pattern lets you skip the rules of the puzzle
B. The pattern predicts the answer for towers you have never played, like five or six discs
C. Patterns make the discs move by themselves
D. A pattern proves the puzzle is impossible

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**8. Before solving, the lesson asked you to DESCRIBE what the solution will look like. Why is imagining the ending a useful way to start a puzzle?**

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**9. Will the fewest number of steps for a Tower of Hanoi puzzle always be an ODD number?**

A. Yes — doubling any whole number gives an even number, and adding 1 makes it odd
B. No — every other tower size gives an even number of steps
C. Yes — but only because 7 happens to be odd
D. No — the steps alternate between odd and even

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**10. Use the doubling pattern to complete the table of fewest moves.**

| Discs | Pattern: double the previous, add 1 | Fewest moves |
|---|---|---|
| 3 | 2 × 3 + 1 | |
| 4 | 2 × 7 + 1 | |
| 6 | 2 × 31 + 1 | |